Development of an AI-Based Diagnostic Method for Malaria Diagnosis from Thick Blood Smears

Yassine Hermak
Supervisor: Prof. Dr.-Ing. A. Ibenthal

Progress Update - 10.09.2026

HAWK University of applied sciences and arts - Göttingen

In cooperation with : Ethiopian AI Institute (EAII) & CarByte Engineering GmbH

Updates since last meeting

- LR2 results (bug found and fixed)
- Full ResNet50 resolution comparison - HR, LR1, LR2, LR3
- Explainability Grad-CAM - LR3 only
- Architecture comparison - ResNet50 vs ResNet18 vs MobileNetV3-Small
- Nexts steps

Correction

A bug in the ResNet50-LR2 Test Script (second Lower Resolution 112px)

What happened
ResNet50-LR2 test script still resized to 160px (LR1's setting) instead of 112px

How it was caught
Grad-CAM script (correctly configured) gave false-negative counts that didn't match test.py

Corrected numbers

Accuracy	Sensitivity	Specificity
98,43%	99,60%	94,09%
~~87,83%~~	~~84,53%~~	~~100,0%~~

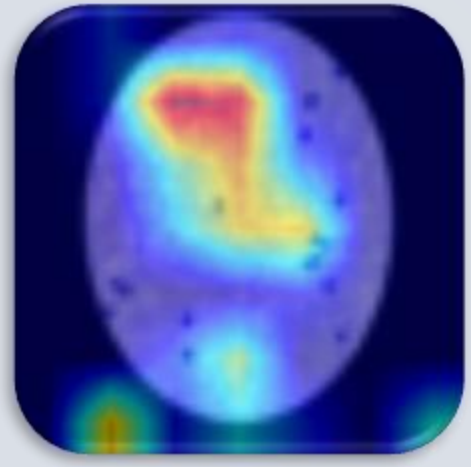
Found and corrected on 20.08.2026

ResNet50 — full resolution comparison

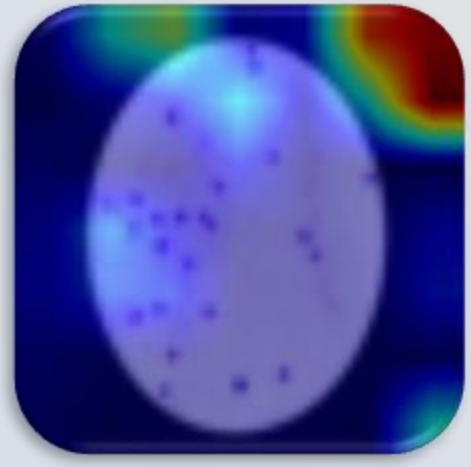
	Accuracy	Sensitivity	Specificity	ROC-AUC
HR (224px)	99.58%	99.73%	**99.01%**	0.9997
LR1(160px)	99.58%	99.87%	**98.52%**	0.9999
LR2(112px)	98.43%	99.60%	**94.09%**	0.9994
LR3(56px)	92.86	98.00%	**73.89%**	0.9698

Sensitivity stays strong across all four levels - specificity is what degrades sharply at LR3

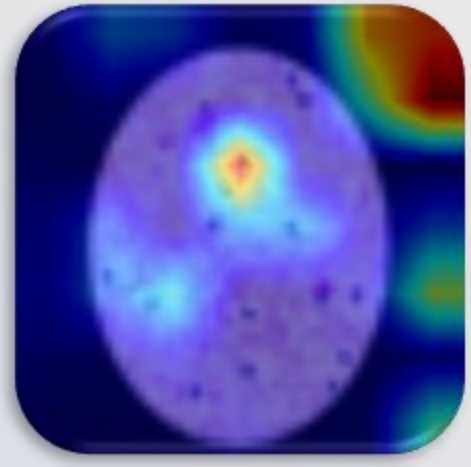
Explainability : Grad-CAM on LR3



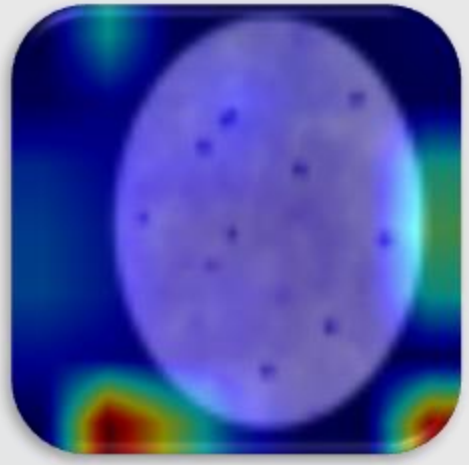
False positive



False negative



True negative



True positive

- **Same pattern in all four.**
- **Strong attention on background/corners, not just the cell, even on the true positive.**

Why this might be happening

Last conv layer
Only a 7*7 grid

→

Stretched to 224*224
imprecize

→

Tiny and blurry parasite
Weak signal at ResNet50-LR3

Documented limitation of Grad-CAM, not specific to my model.
Selvaraju et al. (2017), the original Grad-CAM paper, notes the heatmap resolution is limited by the last convolutional layer's coarse spatial grid

Planned verification:
Comparison against HR Grad-CAM, and trying Grad-CAM++.

Moved on to ResNet18/MobileNet training in parallel rather than being stuck.

Architecture comparison - HR vs LR3

	Resolution	Accuracy	Sensitivity	Specificity	ROC-AUC
ResNet50	HR (224px)	99.58%	99.73%	99.01%	0.9997
ResNet50	LR3(56px)	92.86%	98.00%	**73.89%**	0.9698
ResNet18	HR (224px)	99.69%	100.00%	**98.52%**	1.0000
ResNet18	LR3(56px)	91.08%	94.53%	**78.33%**	0.9666
MobilNetV3-S	HR (224px)	99.69%	100.00%	**98.52%**	0.9999
MobilNetV3-S	LR3(56px)	**89.30**	94.80%	**68.97%**	0.9378

All three tied at HR. At LR3, MobileNet is worst on every metric. ResNet50 keeps the best accuracy and sensitivity, but ResNet18 actually edges it out on specificity.

What this means

Deployment tension
MobileNet is the lightest, most realistic model for a smartphone. But it's also the one that degrades the most at low resolution.

ResNet50 vs ResNet18: Open question
ResNet50 wins on sensitivity and accuracy, ResNet18 wins on specificity at LR. No clear explanation yet for why.

Core thesis tradeoff
model size (deployability) vs. robustness to resolution loss.

Augmentation applies only to training

Next steps

1. **Benchmarking against literature**
   compare my results against published numbers from the review papers

2. **Continue thesis writing**
   methodology and results chapters for everything done so far

3. **Extend explainability (Grad-Cam) to ResNet18 / MobileNet**
   needed to confirm whether the edge-attention pattern is ResNet50-specific or general

Thank you for your attention

Yassine Hermak